CAPSTONE TRACK A · WEEK 1 REVIEW

ChartAudit-Fin v2.0

A blind-labelled finance and economics chart dataset for fine-tuning Qwen2.5-VL-3B

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WEEK 1 FEEDBACK

The labeller must never see the correct answer

The concern

In v1 the teacher model wrote evidence and reasoning while it could see the ChartQA answer. Its reasoning might just justify the answer instead of reading the chart.

What v2.0 does

Rebuilt from raw ChartQA. Items chosen by written rules only, then labelled blind, then compared with the human answers. Disagreements are dropped.

V2.0 AT A GLANCE

Finance and economics charts, labelled blind

1,033
train items

124
val items

180
test items, human answers, never labelled

0
charts shared between splits

Each record: chart image + question → JSON with evidence, rationale, answer type, unit, answer. Plus confidence, domain and domain tags. Two inputs: image only, and image + data table.

THE PIPELINE

Five steps from raw ChartQA to final files

1

Rules

Select items from raw ChartQA. No model.

2

Hide answers

Packs hold image + question. Answers locked in a key.

3

Blind label

Evidence, rationale, answer, confidence.

4

Compare

Open the key. Match? Keep. Disagree? Drop.

5

Final files

Blind evidence + human answer.

Scripts: select_clean.py → make_packs.py + leak_check.py → labeller → compare.py → build_final.py

STEP 1 · SELECTION BY RULES

Seven written rules, no model, fixed seed

Rule	Problem it solves
1 Basic validity	No table, fragment question, empty answer
2 De-duplication	Same question counted twice
3 Leakage control	Same table in train and test
4 Consistency	Wrong gold (0.119 for 11.9%), unclear colours
5 Domain filter	Keep finance and economy
6 Stratified sample	Balanced mix, reproducible
7 Unanswerable items	Teach the model to refuse

Train funnel

7,398 human questions

6,595 pass rules 1–4

2,306 finance / economy

1,155 selected (55 unanswerable)

1,033 kept after blind check

STEPS 2–4 · BLIND LABELLING

How we know the labeller was blind

Split the data

Packs hold only a neutral image name (q0001.png) and the question. Answers sit in a separate key folder.

Leak check

A script fails if any pack line holds more than pack_id, image and question.

Fresh labeller

A new Claude Sonnet per batch of 50, with a saved written prompt. The key is opened only after labelling.

Compare: exact match = **high** · exact but labeller unsure = **medium** · anything else **low**, dropped

BLIND CHECK RESULTS

9 in 10 blind answers match the human answer

	Train	Val
Labelled blind	1,155	137
High (exact match)	1,018 (88.1%)	121 (88.3%)
Medium (exact, labeller unsure)	15	3
Low (dropped)	122 (10.6%)	13 (9.5%)
Kept	1,033	124

Why disagreements happen

- 75 of 122 are arithmetic
- ChartQA stores 46% as 0.46
- Ambiguous questions
- Colour names (navy vs dark blue)
- Gridline readings (7.5 vs 7.4)

REAL EXAMPLES FROM TRAIN

High, medium and low in practice

High · kept

“By how much value is 2023 higher than 2016?”

Human 1528 · Blind 1528
“4658 − 3130 = 1528”

Medium · kept

“Sum of cut back and keep same attitudes in 2012?”

Human 83 · Blind 83
Exact, but no point at 2012, so the labeller marked itself unsure

Low · dropped

“Difference between good and bad influence on politics?”

Human 0.06 · Blind 6 (%)
“36 − 30 = 6”: ChartQA stored 6% as 0.06

Also dropped: Belgium, human 7.4 vs blind 7.5 read from gridlines. And a ratio “40:49” saved in ChartQA as a clock time, 1.7007. Many low items are errors in ChartQA itself.

SECOND TEACHER · GPT

29/30

Two model families agree on the same 29

30 random train items, same prompt, answers never shown (ChatGPT and Copilot). Claude matched the same 29. Both missed one ambiguous chart, already dropped.

Details: blind_labelling/runs/gpt_check/RESULTS.md

OTHER DOMAINS KEPT, NOT DELETED

Every domain is counted in a catalogue

Domain	Train	Val	Test
Finance / economy	**2,306**	**309**	**398**
Health	433	78	88
Business	248	23	46
Other	3,608	458	620
Total	**6,595**	**868**	**1,152**

all_domains_catalogue.jsonl lists every question that passes rules 1–4, with domain tags and a single domain. No labels.

Per-tag train counts overlap: finance 1,423, economy 1,213, business 865, health 553.

BASELINES · QWEN2.5-VL-3B, BEFORE FINE-TUNING

Two examples help a little. Step-by-step reasoning does not.

Zero-shot

52.8%

95 of 180 · chart + question only

Few-shot (2) · best

57.2%

103 of 180 · two solved train examples first

Chain of thought

52.2%

94 of 180 · evidence and reasoning first

	Zero-shot	Few-shot	CoT
Arithmetic / comparison (96)	43.8%	42.7%	**49.0%**
Retrieval (35)	74.3%	**82.9%**	68.6%
Yes / no (10)	0 of 10	**3 of 10**	2 of 10
Unanswerable, refused (9)	4 of 9	**7 of 9**	6 of 9
Output matches the schema	98.9%	98.9%	**51.1%**

Relaxed accuracy: numbers within 5% of the human answer, text after normalising, refusals correct when the answer is null. Image only, bf16, greedy decoding, one SLURM job on an L40S.

WHAT ZERO-SHOT GETS WRONG

It reads charts, but ignores the answer format

Zero-shot, by type	Accuracy
Retrieval (look-up)	74.3%
Number answers	63.1%
Arithmetic / comparison	43.8%
Unanswerable (refused)	4 of 9
Text answers	35.9%
Yes / no	**0 of 10**

Yes / no: it answers a number

“Is the total of the two bar values greater than 80?”

Human: **No** · Model: **76.3**

It did the sum, then never answered the question. Every training target shows the right answer type, so fine-tuning should fix this.

LIMITS AND NEXT STEPS

What is left before fine-tuning

Limits

- Isolation rests on packs, leak check and saved prompt, not a sandbox
- ChartQA is public: a large model may have seen it
- Dropping disagreements removes some hard training items

Next

- Bar to beat: 57.2% (few-shot), plus yes/no and the evidence list format
- Three-reviewer human audit of 50 items, Fleiss’ kappa
- Merge the v2.0 pull request and freeze the dataset
- Week 2: LoRA fine-tune on 1,033 train, compare on test

Final submission 27 November · method and counts: data/v2.0/README.md